Angles on Parallel Lines: Harder

Name _____ **Date** _____ No calculator **Version 1**

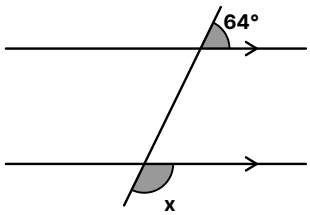
KEY WORDS parallel · transversal · corresponding · alternate · co-interior · parallelogram · trapezium · reasons

A Two steps ●○○ ANSWER ON THE SHEET

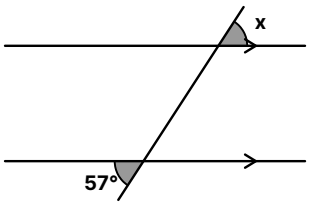
Remember: Sometimes no single rule links the two angles. Find a third angle first, then give both reasons.

1. The arrows show that the lines are parallel. Find the size of the angle marked x. Give both reasons.

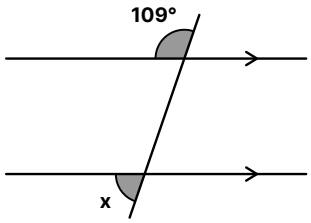
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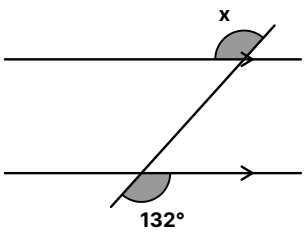
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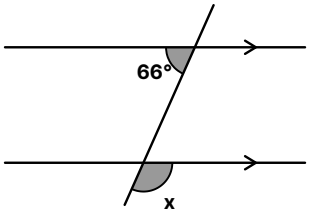
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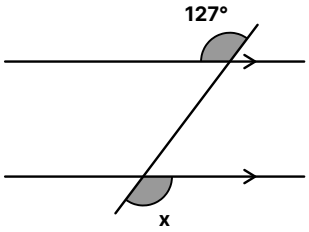
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e)



f)

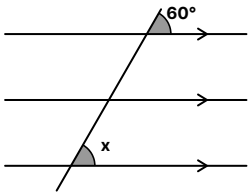


B Three parallel lines ●●○ ANSWER ON THE SHEET

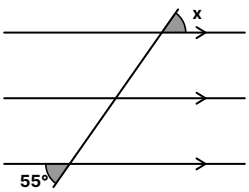
Remember: The rules work between any two of the parallel lines, not only neighbours: corresponding angles are equal at every crossing.

2. All three lines with arrows are parallel. Find the size of the angle marked x.

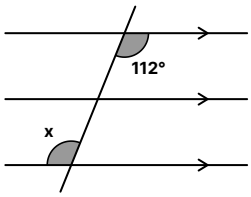
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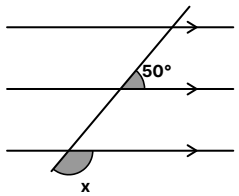
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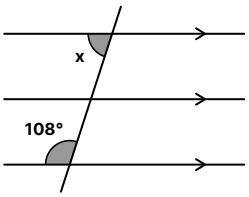
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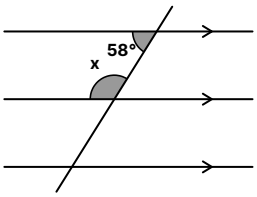
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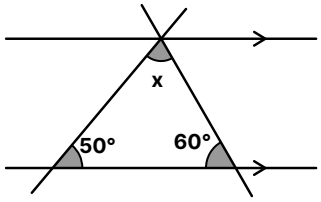
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C A triangle between the lines ●●● WORK IN YOUR BOOK

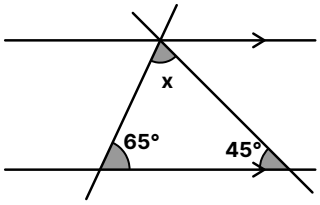
Remember: Two lines crossing the parallels make a triangle. Use alternate angles to move an angle up to the top line, then angles on a straight line, or use the angle sum of a triangle.

3. The two lines with arrows are parallel. Find the angles marked with letters. Give reasons.

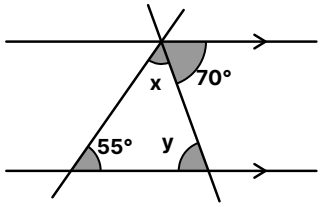
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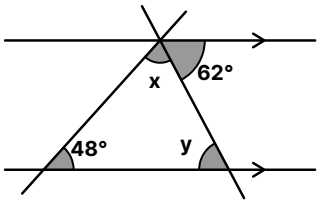
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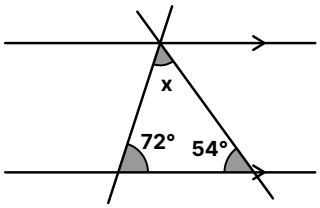
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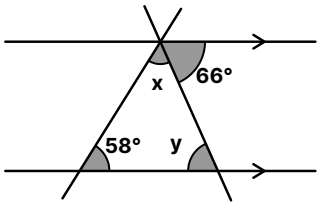
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e)



f)

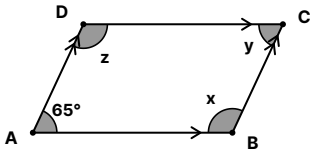


D Parallelograms and trapeziums ●●● WORK IN YOUR BOOK

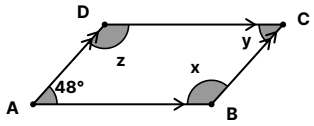
Remember: A parallelogram has two pairs of parallel sides, so angles next to each other are co-interior and add to 180°. A trapezium has one pair.

4. Find the angles marked with letters. Give reasons.

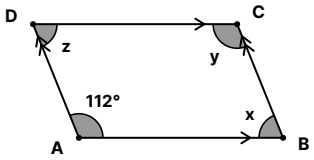
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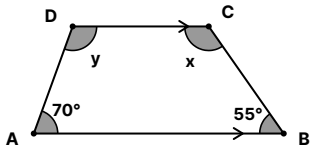
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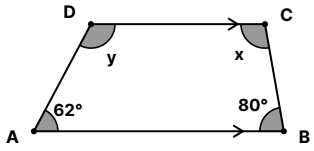
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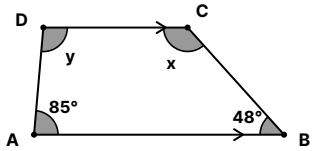
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e)



f)

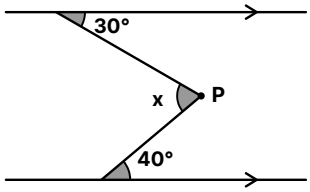


E The zigzag ●●● WORK IN YOUR BOOK

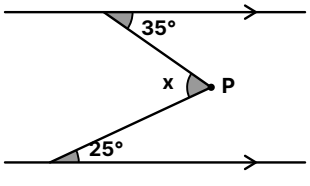
Remember: When a point sits between the parallel lines, draw a third parallel line through it. The angle at the point splits into two alternate angles.

5. The two lines with arrows are parallel. Find the angle marked x.

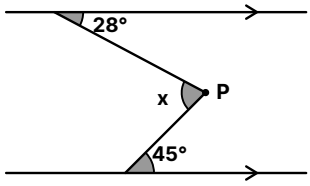
a)



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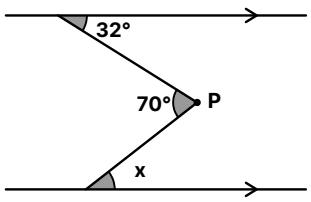


c)

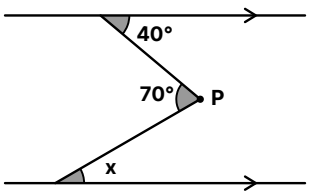


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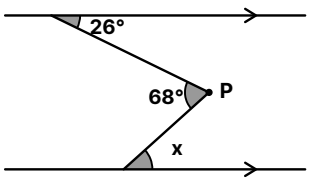
d)



e)



f)



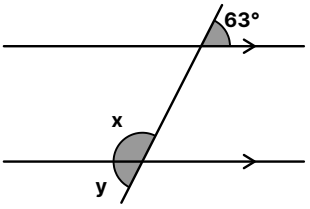
F Two unknowns, with reasons ●●●

WORK IN YOUR BOOK

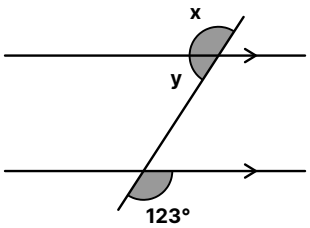
⚠ **Watch out:** Write a reason for each angle, in the order you found them: a reason is only valid if the angle it uses is already known.

6. Find x and y. Give a reason for each.

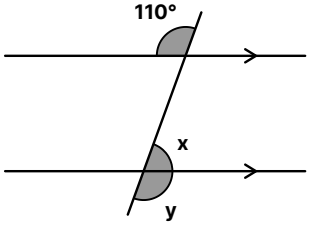
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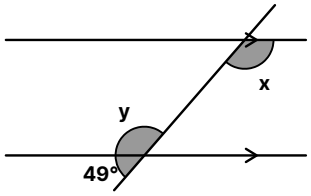
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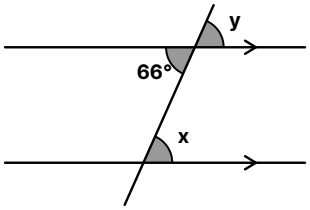
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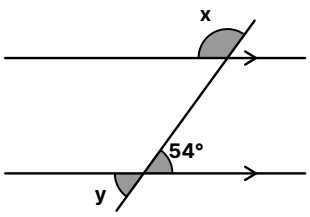
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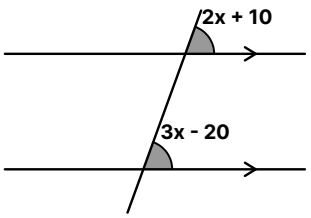


Extension

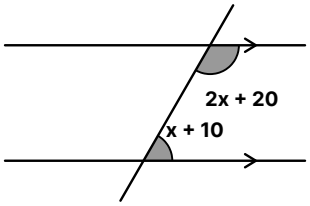
FOR EARLY FINISHERS

7. Both angles are given in terms of x. Write an equation, solve it, and find the size of each angle.

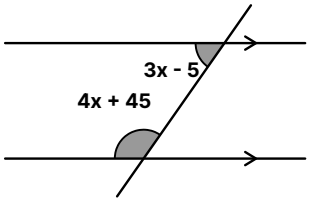
a)



b)



c)



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Angles on Parallel Lines: Harder

Name _____ **Date** _____ No calculator **Version 2**

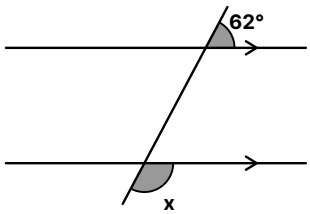
KEY WORDS parallel · transversal · corresponding · alternate · co-interior · parallelogram · trapezium · reasons

A Two steps ●○○ ANSWER ON THE SHEET

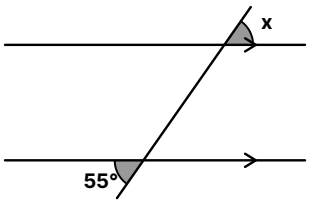
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1. The arrows show that the lines are parallel. Find the size of the angle marked x. Give both reasons.

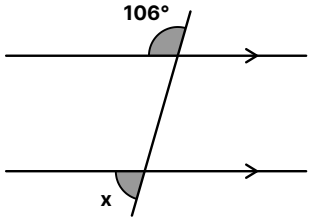
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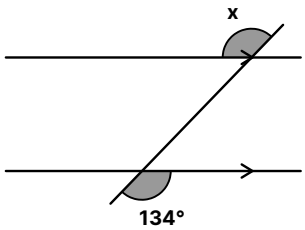
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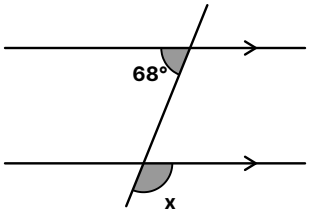
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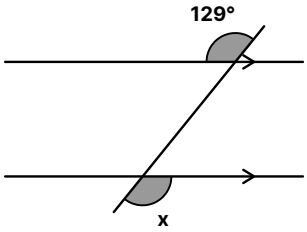
d)



e)



f)

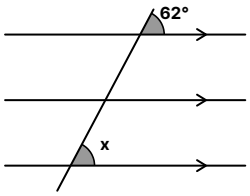


B Three parallel lines ●●○ ANSWER ON THE SHEET

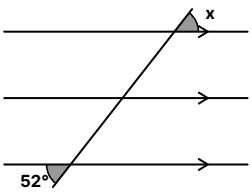
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2. All three lines with arrows are parallel. Find the size of the angle marked x.

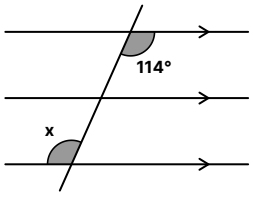
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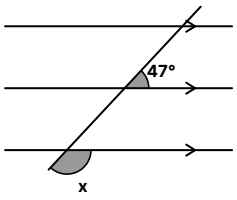
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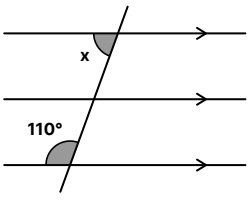
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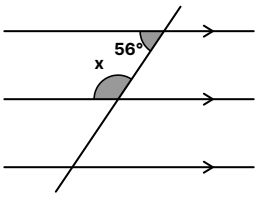
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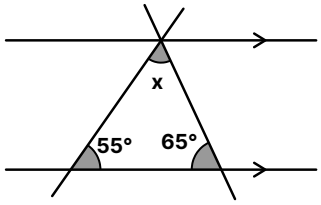
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C A triangle between the lines ●●● WORK IN YOUR BOOK

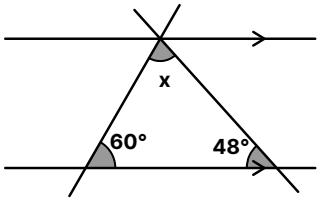
Remember: Two lines crossing the parallels make a triangle. Use alternate angles to move an angle up to the top line, then angles on a straight line, or use the angle sum of a triangle.

3. The two lines with arrows are parallel. Find the angles marked with letters. Give reasons.

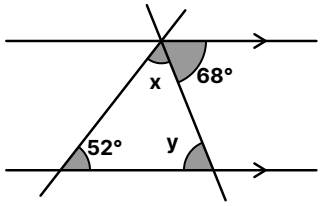
a)



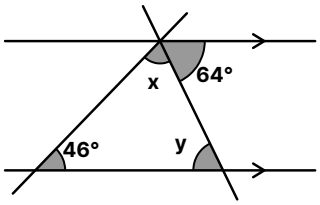
b)



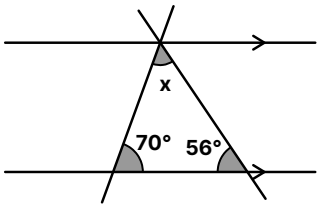
c)



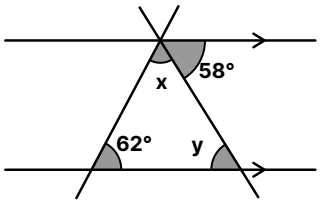
d)



e)



f)

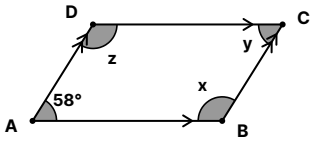


D Parallelograms and trapeziums ●●● WORK IN YOUR BOOK

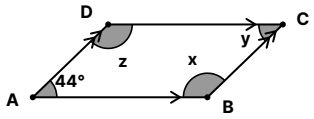
Remember: A parallelogram has two pairs of parallel sides, so angles next to each other are co-interior and add to 180°. A trapezium has one pair.

4. Find the angles marked with letters. Give reasons.

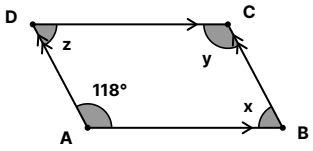
a)



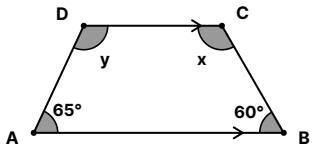
b)



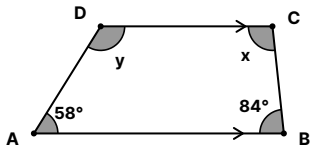
c)



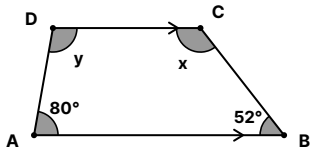
d)



e)



f)

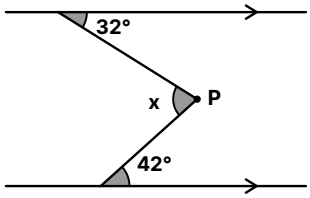


E The zigzag ●●● WORK IN YOUR BOOK

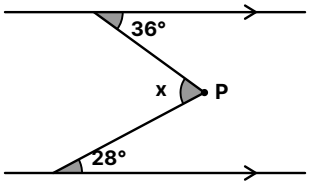
Remember: When a point sits between the parallel lines, draw a third parallel line through it. The angle at the point splits into two alternate angles.

5. The two lines with arrows are parallel. Find the angle marked x.

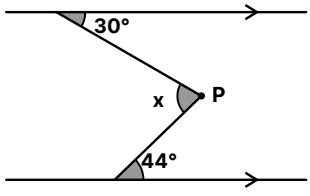
a)



b)

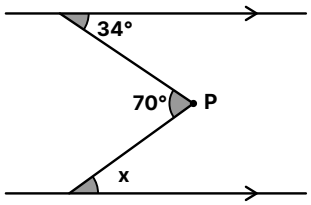


c)

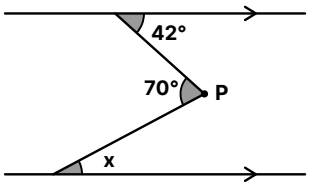


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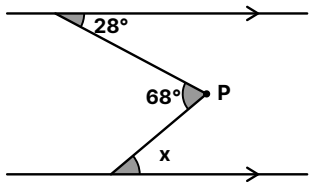
d)



e)



f)



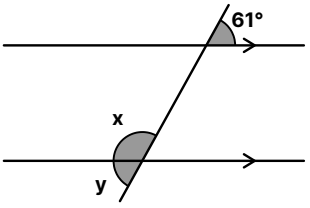
F Two unknowns, with reasons ●●●

WORK IN YOUR BOOK

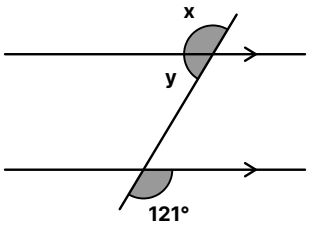
⚠ **Watch out:** Write a reason for each angle, in the order you found them: a reason is only valid if the angle it uses is already known.

6. Find x and y. Give a reason for each.

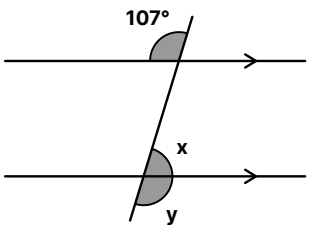
a)



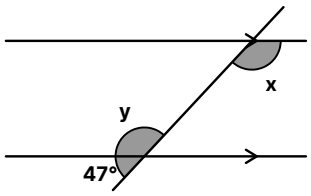
b)



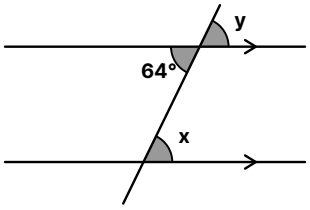
c)



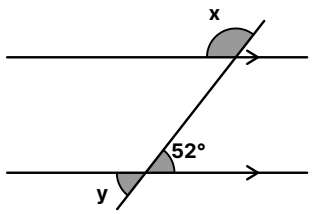
d)



e)



f)

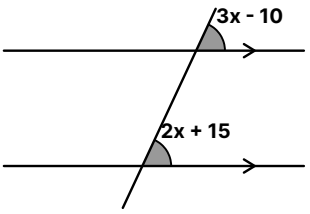


Extension

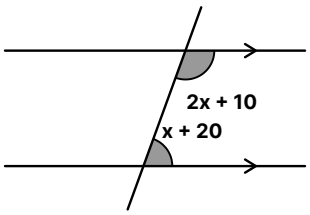
FOR EARLY FINISHERS

7. Both angles are given in terms of x. Write an equation, solve it, and find the size of each angle.

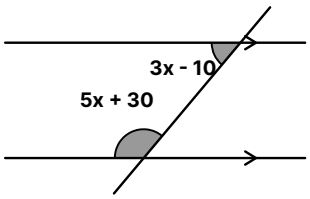
a)



b)



c)



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